Quantum Circuit Learning With Parameterized Boson Sampling

Jinjing Shi, Yongze Tang, Yuhu Lu, Yanyan Feng,
Ronghua Shi, and Shichao Zhang, *Senior Member, IEEE*

Abstract—A quantum circuit learning approach is studied to carry out the fast-fitting of Gaussian functions. First, a parameterized structure is designed for quantum circuits based on the boson sampling model. And then, the training procedure of exploiting gradient-based optimizations is presented to iteratively update the gradient of the loss function concerning circuit parameters. For efficiency, two kinds of circuit loss, the kernel maximum mean discrepancy and the mean absolute error, are used in the training procedure, which are both competent to achieve quantum circuit learning well. It is significant that the two circuit losses assist in reducing the variance to 2.54×10^{-6} and 6.91×10^{-6}, respectively. Finally, a kind of quantum circuit fixed structure is developed with the boson sampling model that can decrease the model complexity as the circuit depth d grows. Sets of experiments have been conducted to evaluate the proposed quantum circuit learning scheme, and demonstrate that our parameterized approach is efficient and promising, and it is worth looking forward to solving practical application problems with quantum computers since valid quantum circuits for Gaussian function fast-fitting can be designed indeed.

Index Terms—Quantum computing, hybrid quantum-classical approach, quantum circuit learning, boson sampling, parameterized boson sampling, parameterized quantum circuit, gaussian function fitting

1 Introduction

Quantum computing [1], [2] is a new computing model that follows the principles of quantum mechanics to regulate quantum information units for computation. Quantum computers can perform specific tasks with exponential computational speedup using quantum algorithms compared to classical computers. For example, Shor's algorithm can achieve exponential acceleration compared with classical algorithms for large integer factorization, which may pose a threat to RSA public key cryptosystem [3]. Another important quantum algorithm Grover's algorithm [4] is capable to search a set of unordered data more efficiently in unstructured search. In recent years, more and more quantum algorithms have been proposed to demonstrate the advantages of quantum computing in various fields, such as machine learning [5], ridge regression [6], visual tracking [7], and association rules mining [8]. With the development of quantum technology, the hybrid quantum-classical (HQC) approach [9] is starting to gain prominence in research, and it is capable to solve practical problems by using both quantum and classical resources. For example, parameterized quantum circuit (PQC) [5], a kind of HQC system that can be considered as a machine learning model, has attracted a lot of attention. By optimizing a loss function on the parameter vectors to reduce it until convergence, PQC can be used to train quantum circuits to solve some specific application tasks, such as supervised learning tasks [10], [11], e.g., classification and regression, generative modeling tasks [12], and quantum algorithm learning [13]. PQC can outperform any classical neural network for generative tasks [14] and efficiently simulate probability distributions generated by instantaneous quantum polynomial-time (IQP) circuit [15] because of its expressive power, while it is difficult for classical neural networks. It can offer a way to demonstrate quantum supremacy in the noisy intermediate-scale quantum (NISQ) era [5].

On the other hand, many scholars have lately explored the practicality of achieving a large enough quantum computation that definitively demonstrates quantum supremacy [16], and boson sampling (BS) [17] is considered as a strong candidate towards an experimental demonstration of quantum algorithmic supremacy. The BS problem that samples random outcomes from the probability distribution induced by the collection of indistinguishable photons evolving in a linear optical network [17], [18], [19], cannot be efficiently simulated on a classical computer. Since some reports have concluded that large-scale BS has the potential to challenge the extended Church-Turing thesis [20], [21], [22], there have been significant efforts to implement it and propose efficient models, such as scattershot BS [23], [24], [25] and Gaussian boson sampling (GBS) [26], [27].

It is encouraging that the quantum computing prototype "jiuzhang" has been invented with GBS [28] in 2020. The photonic quantum computer "jiuzhang" can generate up to 76 output photon clicks in 200s which yields a Hilbert space of 10^{30}, while the TaihuLight (Fugaku) supercomputer using a highly optimized algorithm would be 8×10^{16}s which is 2.5

- *The authors are with the School of Computer Science and Engineering, Central South University, Changsha 410083, China. E-mail: {shijinjing, tangyongze, shirh}@csu.edu.cn, {yuhu_lu, fengyanyanhenu}@163.com, zhangsc@mailbox.gxnu.edu.cn.*

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TABLE 1
Table of Notations

m	The number of modes
n	The number of photons
d	The maximum of layers
U	The unitary transformation of quantum circuit
M	The number of output configurations
$\mathrm{Per}(U_S)$	The permanent of the submatrix U_S
$R(\phi)$	The phase shifter
$BS(\theta, \varphi)$	The beam splitter
$\mathcal{L}$	The loss function
$K(x, y)$	The kernel function in MMD loss
N_1, N_2	The number of phase shifters and beam splitters
η	The learning rate

billion years [28]. There are also a series of applications that apply BS to solve diverse real problems. In 2015, Huh *et al.* [29] displayed that a BS device with a modified input state can be used to generate molecular vibronic spectra, including complicated effects such as Duschinsky rotations. In 2018, Arrazola *et al.* [30] showed that GBS is a useful tool for finding dense subgraphs, which makes it possible to find approximate solutions to the densest k-subgraph problem by constructing a customized stochastic algorithm. In 2019, Huang *et al.* [31] developed a quantum symmetric encryption which is built on BS using multi-photon interference. In 2020, Banchi *et al.* [32] showcased a general formula framework for training GBS distributions by applying it to problems in stochastic optimization and unsupervised machine learning. In 2020, Feng *et al.* [19] also developed an arbitrated quantum signature protocol using BS-based unitary operation encryption, where the BS-based unitary is governed by adjustable parameters [33]. The branches of applications of BS shorten the research process of solving practical problems with BS indeed. While the structure of BS and its execution mode always remain unchanged, which may limit the application scenarios of the BS.

Inspired by the PQC model, the BS model is improved with a parameterized structure designed in this paper. For simplifying the description, it is denoted as parameterized boson sampling (PBS). It can be an implicit clue that the BS model can be parameterized with quantum gate parameters, and the definable parameterized principle of BS is discussed in this work to derive an HQC approach for function fitting [34], [35]. Different from the classical methods in [34], [35], the PBS for function fitting is designed with quantum circuits, which is able to fit a Gaussian distribution with $\binom{m+n-1}{n}$ sample points. The training is executed by exploiting gradient-based optimization algorithms [36], [37] to iteratively update the gradient of the loss function concerning circuit parameters. Since the designed PBS circuit includes the structure of a fixed gate, the number of free circuit parameters is scaled by a polynomial of the mode number. The simulation performed on Strawberry Fields [38] shows that the PBS can efficiently simulate Gaussian function, which implies the wide perspective application of PBS, such as engineering, mathematics, signal/image processing, and industrial processes.

The main contributions of this research are summarized as follows.

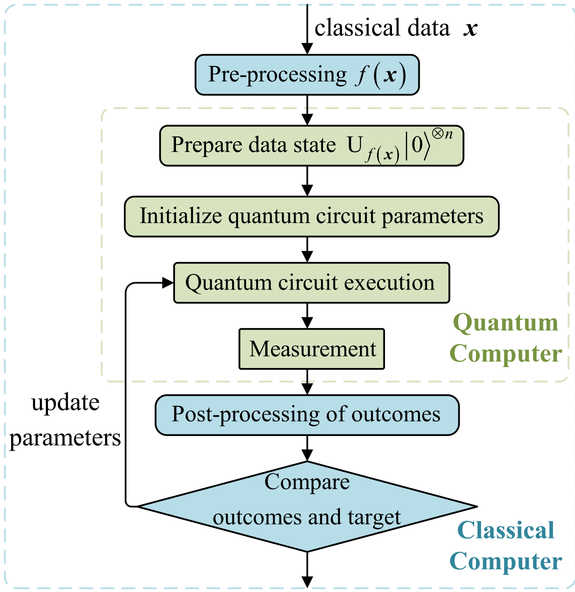


Fig. 1. Hybrid quantum-classical process.

- A parameterized structure is proposed for advancing the BS model to the PBS model. As a variational quantum circuit, the PBS model is useful in data mining and machine learning.
- A quantum circuit fixed structure is developed with the PBS model. It clearly scales down the model complexity that the parameter number of the quantum circuit can be scaled as the polynomial of the mode number.

The rest of this paper is organized as follows. Related work and basic concepts are briefly recalled in Section 2. And then, the PBS circuit learning is proposed in Section 3. In Section 4, experiments are conducted with training examples to show the effectiveness of the PBS circuit learning. Finally, this research is concluded in Section 5.

2 Related Work

In this section, we first review the general HQC approach, then introduce one of the HQC models, the PQC model, and the basics of the boson sampling model which are closely related to ours. At the very beginning, the notations used in the paper are summarized in Table 1.

2.1 Hybrid Quantum-Classical Approach

The hybrid quantum-classical approach is able to make use of the existing quantum device to the greatest extent. The general hybrid quantum-classical algorithm consists of two main components, i.e., the classical computer and the quantum computer, as shown in Fig. 1. First, the data is preprocessed on a classical computer to specify a set of initial parameters for the quantum computer. Second, the quantum computer prepares the initial state, performs the quantum circuit, and implements measurements. Then, the classical computer compares the target data with outcomes after post-processing and updates the circuit parameters to improve the accuracy of outcomes by learning algorithm [39], [40], [41]. Finally, the overall process is run in a closed

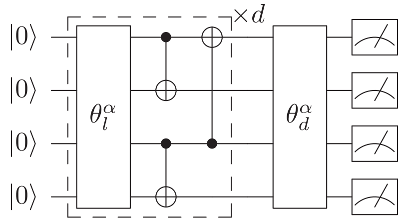


Fig. 2. An quantum circuit example of QCBM.

loop between the classical and quantum computer and the outcomes approach the goal step by step.

Several variants of HQC algorithms have recently been demonstrated and can be applied to tackle small-scale problems in chemistry, combinatorial optimization, and machine learning. For example, the variational quantum eigensolver [42] is used to implement quantum chemical simulation by estimating the ground state energy of molecules in the system [43], [44]. The quantum approximate optimization algorithm has been used for finding approximate solutions of classical Ising model [45] and clustering problems formulated as MaxCut [46].

The application of HQC algorithms in machine learning include the quantum circuit learning [47], [48], the quantum generative adversarial networks [49], the quantum autoencoder for quantum data compression [50] and the variational quantum state eigensolver [51].

For example, as a class of data-driven quantum circuit learning, the quantum circuit born machines (QCBM) [36] can represent the probability distribution of the classical dataset by using the variational quantum circuit. Given a dataset $\mathbf{D} = \{\boldsymbol{x}_1, \ldots, \boldsymbol{x}_D\}$ that are D independently and identically distributed with the target probability distribution, the QCBM can generate samples that approximate the unknown target probability distribution. As shown in Fig. 2, the QCBM takes the product state $|0\rangle^{\otimes n}$ and evolves it to the output state $|\psi_{\boldsymbol{\theta}}\rangle$ by unitary transformation $U(\theta)$. The overall circuit layout includes a series of single-qubit rotation layers and entanglement layers. The rotation layers are parameterized with rotation parameters $\boldsymbol{\theta} = \{\theta_l^\alpha\}$ where the layer index satisfies $l \in [0, d]$ with d as the maximum depth of the circuit. According to the qubit index j, α consists of three arbitrary rotation gate index where the arbitrary rotation gate can be expressed as

$$U(\theta_l^j) = R_z(\theta_l^{j,1}) R_x(\theta_l^{j,2}) R_z(\theta_l^{j,3}), \tag{1}$$

with $R_m(\theta) = \exp(\frac{-i\theta\sigma_m}{2})$. Then, the output state $|\psi_{\boldsymbol{\theta}}\rangle$ is measured on a computation basis to obtain a sample of bits $\boldsymbol{x} \sim p_{\boldsymbol{\theta}}(\boldsymbol{x}) = |\langle \boldsymbol{x}|\psi_{\boldsymbol{\theta}}\rangle|^2$. Finally, the model probability distribution $p_{\boldsymbol{\theta}}(\boldsymbol{x})$ can approach the target distribution by parameter learning [52], [53]. The negative log-likelihood as a cost function is minimized to optimize the parameters,

$$L(\boldsymbol{\theta}) = -\frac{1}{D}\sum_{d}^{D} \ln[\max(\epsilon, P_{\boldsymbol{\theta}}(\boldsymbol{x}_d))], \tag{2}$$

where $\epsilon > 0$ is a small number to avoid singularities of the cost function.

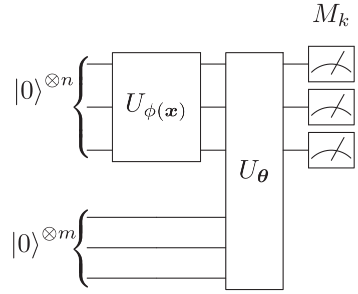


Fig. 3. The parameterized quantum circuits model.

2.2 Parameterized Quantum Circuits

PQC include two basic circuits, one is the encoder circuit $U_{\boldsymbol{\phi}(\boldsymbol{x})}$ that transforms data into qubits, and the other is the variational circuit $U_{\boldsymbol{\theta}}$ that aims to implement task approximations [5], as shown in Fig. 3.

The encoder circuit $U_{\boldsymbol{\phi}(\boldsymbol{x})}$ implement a function of encoding classical data into a quantum state that mapping the classical data to the high-dimensional vector space of the states with n qubits, $\boldsymbol{x} \rightarrow U_{\boldsymbol{\phi}(\boldsymbol{x})}|0\rangle^{\otimes n}$, where ϕ is a pre-processed function defined by user which converts the data into circuit parameters. The variational circuit $U_{\boldsymbol{\theta}}$ uses a 'cheap' circuit design that follows a fixed structure of gates. As the number of qubits increases, the dimension of the vector exponentially grows, but the fixed structure can reduce the model complexity and allow the number of circuit parameters to be scaled to a polynomial of the qubit count. In general, PQCs are composed of a set of adjustable rotation gates and fixed gates, e.g., CNOT gates.

The PQC model can be used to perform data-driven tasks as well as classical neural networks. The circuit learning of task relies on minimizing a loss function $L(\boldsymbol{\theta})$, with respect to the circuit parameter vector $\boldsymbol{\theta}$. Several optimization algorithms [54], [55], [56], [57] can be utilized in the PQC model to update the parameter vector. For example, the gradient-based algorithm applies an iterative method to renew circuit parameters in the direction of a sharp decline

$$\boldsymbol{\theta} \leftarrow \eta \nabla_{\boldsymbol{\theta}} L, \tag{3}$$

where $\nabla_{\boldsymbol{\theta}} L$ is the gradient vector and η is the learning rate. The required gradient vector

$$\nabla_{\boldsymbol{\theta}} L = \left\{\frac{\partial L}{\partial \theta_1}, \frac{\partial L}{\partial \theta_2}, \ldots, \frac{\partial L}{\partial \theta_j}, \ldots\right\}, \tag{4}$$

can be evaluated using the finite difference method

$$\frac{\partial L}{\partial \theta_j} \approx \frac{L(\boldsymbol{\theta} + \Delta \boldsymbol{e}_j) - L(\boldsymbol{\theta} - \Delta \boldsymbol{e}_j)}{2\Delta}, \tag{5}$$

where Δ is a (small) hyperparameter and $\boldsymbol{e}_j$ is the Cartesian unit vector in the jth direction.

2.3 Boson Sampling

Boson sampling, first proposed by Aaronson and Arkhipov [17], was the feasible example of demonstrating that quantum computers can speed up in solving certain well-defined tasks compared to classical computers.

The BS model is a quantum system of n identical photons and m modes, requiring only single-photon sources, passive

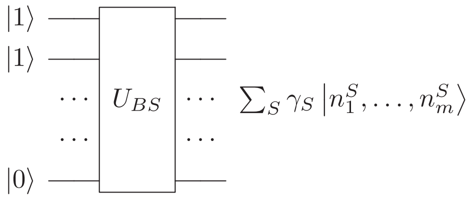


Fig. 4. The boson sampling model.

linear optics (i.e., beam splitters and phase shifters), and photodetection [18], as shown in Fig. 4.

Any computational basis state of BS system [17] can be represented as

$$|\psi\rangle = |n_1, \ldots, n_m\rangle, \tag{6}$$

where n_i represents the photon number in the ith mode, and the photon number satisfies $n_1 + \ldots + n_m = n$ and $n_i \geq 0$. There may appear multiple identical photons in the ith mode at the same time when $n_i > 1$, but the photons can't be created or destroyed in the BS network but are transferred from one mode to another. The form of the output state is the same as one of the input state,

$$|\psi'\rangle = |n'_1, \ldots, n'_m\rangle, \tag{7}$$

where the photon number n'_i also satisfies $n'_i \geq 0$ and $n'_1 + \ldots + n'_m = n$.

According to quantum mechanics, the output state via BS network [18] can be represented as

$$|\psi_{out}\rangle = \sum\nolimits_S \gamma_S |n_1^S, \ldots, n_m^S\rangle, \tag{8}$$

where S denotes one of the different configurations and γ_S is the complex amplitude associated with configuration S. The total number of configurations in the output modes can be denoted as $M = \binom{n+m-1}{n}$. The probability of different photon distributions can be obtained by measuring the output states,

$$\Pr[S] = |\gamma_S|^2, \tag{9}$$

where γ_S satisfies $\sum_S |\gamma_S|^2 = 1$. The configuration amplitude of the output state

$$\gamma_S = \frac{\mathrm{Per}(U_S)}{\sqrt{n_1^S!, \ldots, n_m^S!}}, \tag{10}$$

is related to the permanent $\mathrm{Per}(U_S)$ of the $n \times n$ submatrix U_S, and it is a #P-complete problem [58] which is tough to implement in classical computer.

3 Circuit Learning With Parameterized Boson Sampling

In this section, a Gaussian curve fitting learning scheme based on the PBS model is presented to approximate Gaussian probability distributions from the configuration probability of PBS. We first provide the PBS model and a type of quantum circuit fixed structure of the PBS model. Then, we introduce two loss functions and gradient-based optimization in the learning scheme. At last, quantum circuit learning based on the PBS model is presented.

3.1 Parameterized Boson Sampling

Fig. 5 depicts the schematic of the PBS model. Quite different from the general BS model, the parameterized structure for BS is designed to develop a PBS model where the parameters of passive linear optics are flexible, which can be updated in an iterative manner according to the loss function.

The PBS model treats quantum gates in the BS model as parameterized nodes in machine learning to cope with massive data processing, while adapting the BS model to a learnable model to solve the difficult problems of classical computers in the NISQ era. Just as a classical neuron acquires the upstream input signal and transmits the output signal to the downstream neuron, the front quantum gates in the PBS model transmit photons to the back quantum gates.

In the standard quantum computing, any unitary transform on n qubits can be decomposed into the product of multiple gates, each of which acts on at most two qubits [17]. Similarly, the $m \times m$ unitary transformation U in the PBS model can be also decomposed into the combination of two optical devices, i.e., phase shifter and beam splitter. A phase shifter $R(\phi)$ acts on a single-mode by multiplying the amplitude α_a of the single-mode by $e^{i\phi}$, for the specified angles $\phi \in [0, 2\pi)$, and it acts as the identity on the amplitude of other $m-1$ modes. A beam splitter $BS(\theta, \varphi)$ acting on two modes by multiplying amplitudes α_a and α_b can be expressed as follows, for the specified angles $\theta, \varphi \in [0, 2\pi)$:

$$BS(\theta, \varphi)\begin{pmatrix}\alpha_a \\ \alpha_b\end{pmatrix} = \begin{pmatrix}\cos\theta & -e^{i\varphi}\sin\theta \\ e^{-i\varphi}\sin\theta & \cos\theta\end{pmatrix}\begin{pmatrix}\alpha_a \\ \alpha_b\end{pmatrix}. \tag{11}$$

The angle ϕ can adjust the effect of the phase shifter in the BS model, and θ, φ specify the bias and phase relationship of the beam splitter, respectively [18].

3.2 Quantum Circuit Fixed Structure of PBS Model

We consider the PBS model consisting of alternating phase shifter arrays $R(\boldsymbol{\phi}_i)$ and beam splitter arrays $BS(\boldsymbol{\theta}_i, \boldsymbol{\varphi}_i)$, where the layer index i runs from 1 to d where d is the

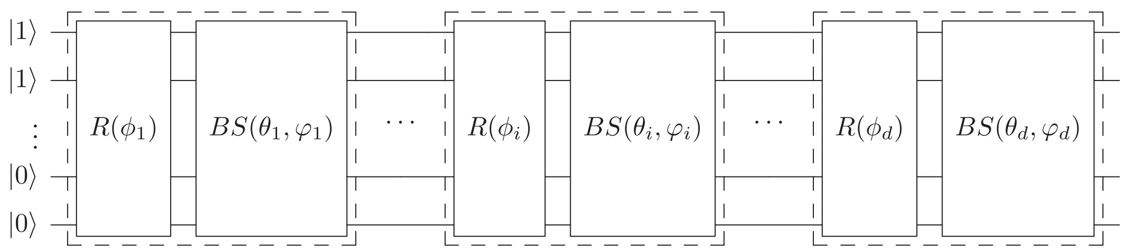


Fig. 5. The parameterized boson sampling model. The quantum circuit has a total of d layers as same as that in Fig. 2, each layer of the quantum circuit consists of a set of $R(\phi_i)$ and $BS(\theta_i, \varphi_i)$, and the d-layer quantum circuit is analogous to the variational circuit in Fig. 3.

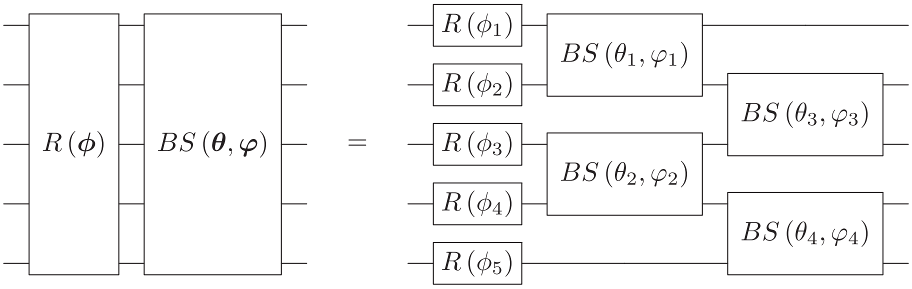


Fig. 6. The quantum circuit fixed structure of PBS model.

maximum depth of the circuit, depicted in Fig. 5. The quantum circuit design is based on a fixed structure that contains alternating blocks of parameterized single-mode rotation and entanglement array. Each block structure begins with parameterized phase shifters $R(\phi_i)$ applied to all modes, followed by two sets of staggered beam splitters $BS(\theta_j, \varphi_j)$, i.e., the entanglement array, illustrated in Fig. 6. Each entanglement array consists of two layers of the beam splitter which act on two adjacent modes according to parity to generate entanglement between different modes.

Each layer of the fixed structure in the PBS quantum circuit is a small-scale interconnected tensor network, which not only enables better representation of quantum states but also reduces the computational cost. Each small-scale tensor network can be simply computed and then the unitary U_{PBS} of the entire PBS quantum circuit can be calculated according to the tensor network. The quantum circuit design of the PBS model follows a fixed structure of gates to reduce the model complexity so that the number of parameters is scaled polynomially in the mode number [5]. If such a PBS circuit acts on m modes, it needs md phase shifters and $(m-1)d$ beam splitters in total.

3.3 Loss Function and Gradient-Based Optimazation

3.3.1 Two Loss Functions

In quantum circuit learning based on PBS model, two evaluation metrics are used as loss functions, i.e., the Maximum Mean Discrepancy (MMD) [59] and the Mean Absolute Error (MAE) [60], [61].

The PBS circuit is trained by employing a two-sample test [62] which determines the distance in the kernel feature space between sample points drawn from the target and model distribution. The MMD is a statistic of the two-sample test problem, that is the difference between the mean function values on two samples. The MMD loss, the optimization objective of this model, can be defined as

$$\begin{aligned}\mathcal{L} &= \left[\mathbf{E}_{\boldsymbol{p},\boldsymbol{p}}[K(\boldsymbol{p},\boldsymbol{p})] - 2\mathbf{E}_{\boldsymbol{p},\boldsymbol{p}'}[K(\boldsymbol{p},\boldsymbol{p}')] + \mathbf{E}_{\boldsymbol{p}',\boldsymbol{p}'}[K(\boldsymbol{p}',\boldsymbol{p}')]\right]^{\frac{1}{2}} \\ &= \left[\frac{1}{M^2}\sum_{i=1}^{M}\sum_{j=1}^{M}K(p_i,p_j) - \frac{2}{M^2}\sum_{i=1}^{M}\sum_{j=1}^{M}K(p_i,p'_j)\right. \\ &\quad \left. + \frac{1}{M^2}\sum_{i=1}^{M}\sum_{j=1}^{M}K(p'_i,p'_j)\right]^{\frac{1}{2}},\end{aligned} \tag{12}$$

where $\boldsymbol{p}, \boldsymbol{p}'$ denote the output probability distribution and the target probability distribution, respectively; M is the sample number of the probability distribution. The Gaussian kernel function $K(x,y) = \exp(-\frac{1}{2\sigma^2}\|x-y\|^2)$ is applied to avoid working in the high-dimensional feature space [36] where σ is bandwidth parameter which controls the width of the Gaussian kernel function. The purpose of the MMD loss is to maximize the probability that the output probability distribution obtained after passing through the PBS network can be close to the target probability distribution. In other words, the smaller the MMD loss is, the better the output probability distribution matches the target [59].

The MAE have been widely used in evaluating the accuracy of a system [63], [64] as another assessment metric for circuit learning in this model, given by

$$\mathcal{L} = \frac{1}{M}\sum_{i=1}^{M}|p_i - p'_i|, \tag{13}$$

where p_i means the prediction rating of the output probability distribution; p'_i represents the true rating in target probability distribution. The MAE is the most natural representation of the average error, which can be visualized in our approach. As same as MMD loss, the smaller the MAE loss is, the more similar the two probability distributions are.

3.3.2 Gradient-Based Optimazation

After evaluating the circuit loss function, any suitable classical optimizer in principle can be used for circuit parameter optimization. In this scheme, we numerically calculate the required gradient for the circuit parameters $\{\boldsymbol{\phi}, \boldsymbol{\theta}, \boldsymbol{\varphi}\}$ using the finite difference method. In the case of $\{\theta_j\}$, the gradient is approximated as

$$\frac{\partial\mathcal{L}}{\partial\theta_j} = \frac{\mathcal{L}(\boldsymbol{\phi}, \boldsymbol{\theta}+\Delta_j, \boldsymbol{\varphi}) - \mathcal{L}(\boldsymbol{\phi}, \boldsymbol{\theta}-\Delta_j, \boldsymbol{\varphi})}{2\Delta_j}, \tag{14}$$

where Δ_j is a small parameter in the jth direction, then the circuit parameter θ_j is updated with

$$\theta_j \leftarrow \theta_j - \eta\frac{\partial\mathcal{L}}{\partial\theta_j}, \tag{15}$$

where η is the adaptive learning rate which controls the rate of the circuit update.

3.4 Quantum Circuit Learning Based on PBS Model

3.4.1 Circuit Learning Framework

Quantum circuit learning based on the PBS model is a hybrid framework where the quantum computer performs

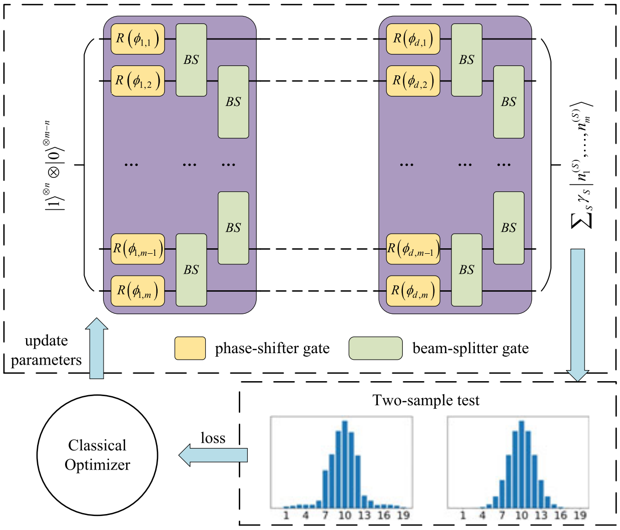


Fig. 7. Quantum circuit learning framework based on PBS model. The input state which is a product state of vacuum and single-photon Fock states is evolved via PBS to the output state. Compare the probabilities obtained from the output state with the target probabilities to get the loss function, and update the quantum circuit parameters according to the loss function.

state preparation, circuit execution, and measurement, and the classical computer evaluates the loss function and updates the circuit parameters, depicted in Fig. 7.

The PBS takes the product states of $|1\rangle$ and $|0\rangle$ as an input, which can be also encoded with the input data by users, and then the input evolves as a composition of different outcomes by a unitary transformation U. The training of the PBS circuit begins with the circuit parameters at random, then the probability of different output configurations obtained with photodetection at the end of the circuit is reorganized to enable subsequent iterative updates. The defined circuit loss function calculates the differences between two probability distributions and the circuit parameters can be updated with gradient-based optimization until the circuit loss function converges.

3.4.2 *Circuit Learning Algorithm*

Algorithm 1 illustrates the pseudocode of the learning algorithm, wherein $|\psi_{in}\rangle$ denotes the input state encoded by the user; $\phi_{i,0}, \theta_{j,0}, \varphi_{j,0}$ indicate the initial parameters of the PBS circuit; L, M mean the maximum number of iterations and the sample number of output configuration probabilities; N_1, N_2 represent the number of phase shifters and beam splitters, respectively; η stands for the adaptive learning rate in the algorithm. We stipulate that $\boldsymbol{p}'$ is the objective Gaussian function, denoted as

$$\boldsymbol{p}' = \{p'_1, p'_2, \ldots, p'_M\}. \tag{16}$$

The probability distribution and circuit loss at the kth iteration are expressed as $\boldsymbol{p}_k$ and $\mathcal{L}_k(\boldsymbol{\phi}, \boldsymbol{\theta}, \boldsymbol{\varphi})$, respectively.

The learning algorithm starts with a random initial guess of the circuit parameters $\{\phi_{i,0}, \theta_{j,0}, \varphi_{j,0}\}$. The output state $|\psi_{out}\rangle$ evolves from the input state $|\psi_{in}\rangle$, where all n single-photons are incident on the m-mode passive linear interferometer described by the parameterized unitary U. The probability distribution $\boldsymbol{p}_k$ can be obtained from $|\psi_{out}\rangle$ and reconstructed as $\hat{\boldsymbol{p}}_k$ by reordering according to the target probability distribution. The circuit loss between the two probability distributions is calculated in accordance with Eqs. (12) or (13), and then the circuit parameters are updated by differentiating the circuit loss. Finally, the entire learning process is iterated until the circuit loss converges.

Algorithm 1. Learning of Parameterized Boson Sampling

Input: $|\psi_{in}\rangle, \phi_{i,0}, \theta_{j,0}, \varphi_{j,0}, L, M, N_1, N_2, \eta, \boldsymbol{p}'$
Output: $\hat{p}_L$

for $k < L$ **do**
  $U = U[R(\boldsymbol{\phi}), BS(\boldsymbol{\theta}, \boldsymbol{\varphi})]$;
  $|\psi_{out}\rangle = U|\psi_{in}\rangle = \sum_l \gamma_{l,k}|n_1^l, n_2^l, \ldots, n_m^l\rangle$;
  for $l < M$ **do**
    $p_{l,k} = |\gamma_{l,k}|^2$;
  end for
  $\boldsymbol{p}_k = \{p_{1,k}, p_{2,k}, \ldots, p_{M,k}\}$;
  if $k = 1$ **then**
    $\hat{\boldsymbol{p}}_k = f(\boldsymbol{p}_k)$;
  else
    $\hat{\boldsymbol{p}}_k \leftarrow \hat{\boldsymbol{p}}_{k-1}$;
  end if
  Compute $\mathcal{L}_k(\boldsymbol{\phi}, \boldsymbol{\theta}, \boldsymbol{\varphi})$ using Eqs. (12) or (13)
  for $i < N_1$ **do**
    $\Delta\phi_{i,k} = \frac{\partial \mathcal{L}_k(\boldsymbol{\phi}, \boldsymbol{\theta}, \boldsymbol{\varphi})}{\partial \phi_{i,k}}$;
    $\phi_{i,k+1} = \phi_{i,k} - \eta\Delta\phi_{i,k}$;
  end for
  for $j < N_2$ **do**
    $\Delta\theta_{j,k} = \frac{\partial \mathcal{L}_k(\boldsymbol{\phi}, \boldsymbol{\theta}, \boldsymbol{\varphi})}{\partial \theta_{j,k}}$;
    $\theta_{j,k+1} = \theta_{j,k} - \eta\Delta\theta_{j,k}$;
    $\Delta\varphi_{j,k} = \frac{\partial \mathcal{L}_k(\boldsymbol{\phi}, \boldsymbol{\theta}, \boldsymbol{\varphi})}{\partial \varphi_{j,k}}$;
    $\varphi_{j,k+1} = \varphi_{j,k} - \eta\Delta\varphi_{j,k}$;
  end for
end for

Calculating the matrix permanents is known to be #P-complete, even harder than NP-complete, and the best-known algorithm requires $O(2^n n^2)$ runtime [18]. The total number of parameters of the quantum circuit is

$$N = N_1 + N_2 = md + 2(m-1)d = (3m-1)d, \tag{17}$$

so the theoretical time complexity of Algorithm 1 can be described as

$$T = NLO(2^n n^2) = O\big((3m-1)2^{n+1}n^2 dL\big), \tag{18}$$

where m, n denote the number of modes and photons, d indicates the quantum circuit layers, and L means the number of iterations.

4 Experiment and Discussion

4.1 Experimental Setup

In this section, we first use numerical simulation to train this PBS circuit with m = 4 and n = 3 on a classical computer to estimate how well the probability distribution can be learned into a Gaussian distribution $\boldsymbol{p}'$. For simplicity, we limit the circuit depth from $d = 2$ to $d = 4$. The input state is $|\psi_{in}\rangle = |1, 1, 0, 1\rangle$ that only the third mode is a vacuum, other three modes all have a photon. The PBS circuit with $m = 4$ and $d = 2$ is constructed with N_1 = 8 phase shifters and N_2 = 6 beam splitters illustrated in Fig. 8 and trained

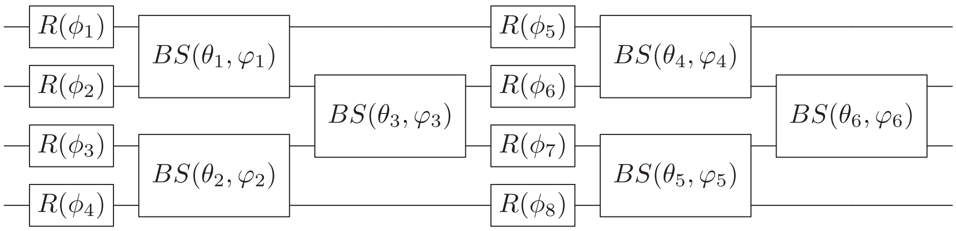
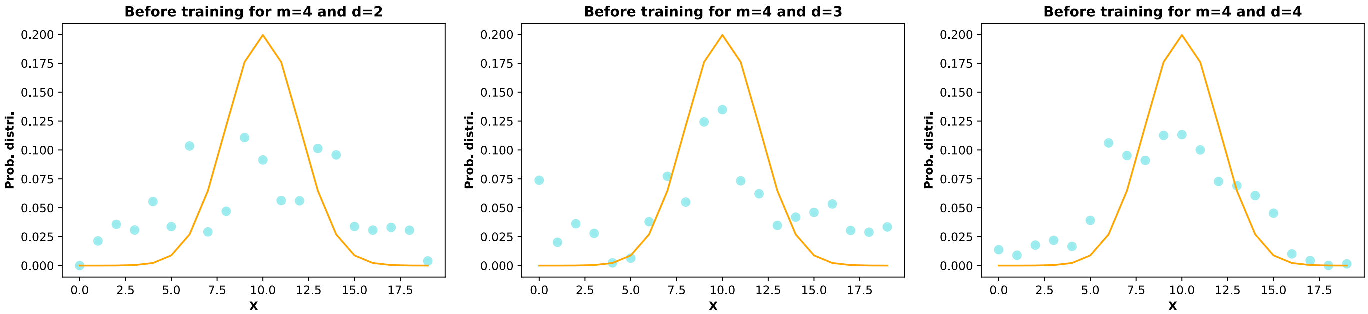
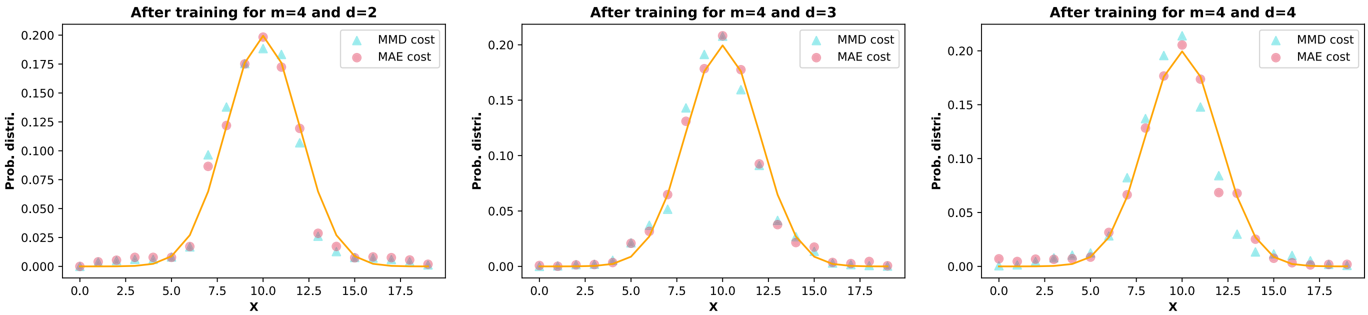


Fig. 8. The PBS circuit with $m = 4$, $n = 3$ and $d = 2$. The PBS circuit consists of 2 layer fixed structures with a total of 8 phase shifters and 6 beam splitters.



(a) Scatterplot before training at different depths



(b) Scatterplot after training at different depths

Fig. 9. Scatterplot for samples generated by the PBS model with $m = 4$.

from a ramdom initialization for $\{\phi_{i,0}, \theta_{j,0}, \varphi_{j,0}\}$. Whether the circuit loss is the MMD loss or the MAE loss, the circuit learning starts with the same initialization for the circuit parameters $\{\phi_{i,0}, \theta_{j,0}, \varphi_{j,0}\}$. We use the Strawberry Fields [38], a cross-platform Python library for photonic quantum computing, to simulate the linear optical network of PBS in order to obtain the probabilities of all possible output configurations. The target Gaussian probability distribution $\boldsymbol{p}'$ is generated with the python statistics library Scipy [65], with mean and standard deviation of 10 and 2, as shown by the orange line in Fig. 9. The number of configurations for PBS with the number of modes $m = 4$ and photons $n = 3$ is the same as the number of sample points, i.e., $M = \binom{m+n-1}{n} = 20$.

Then, we perform further experiments that the mode of the PBS is expanded to $m = 6$. The input photons are still 3 identical photons and placed on the first, second, and fourth modes. The target Gaussian probability distribution $\boldsymbol{p}'$ is also generated with the python statistics library Scipy that the means are both halves of the number of samples and the standard deviations are 4 and 5 for $m = 5$ and $m = 6$, respectively, shown in Table 2. Unlike the experiment with $m = 4$, the extended PBS simulation uses the product of tensor matrics according to Strawberry Fields. The number of configurations for PBS with the number of modes $m = 5$ and $m = 6$ are $\binom{5+3-1}{3} = 35$ and $\binom{6+3-1}{3} = 56$, respectively. The bandwidth of Gaussian kernels used in the MMD loss is $\sigma = \frac{\sqrt{2}}{2}$.

4.2 Experimental Results and Discussion

Fig. 9 shows the probability distributions with $m = 4$, which are composed of the probabilities of all possible output configurations. The probability distributions before training are all chaotic and uncorrelated with the target Gaussian distribution in Fig. 9a, and the probability distributions after 300 training steps fit the target Gaussian distribution well for both the MMD loss and the MAE loss regardless of the depth, depicted in Fig. 9b. The sample points on both sides of the probability distribution after 300 training steps coincide with the target Gaussian distribution, and the sample points in the middle are also highly closed to the Gaussian distribution.

The probability distributions after 300 training steps for $m = 5$ is shown in Fig. 10. For circuit depth from $d = 2$ to $d = 4$, the probability distributions based on MMD loss and

TABLE 2
The Target Distribution Setup

modes	sample number	mean	standard deviation
$m = 4$	20	10	2
$m = 5$	35	17	4
$m = 6$	56	28	5

MAE loss are able to fit the target Gaussian distribution efficiently. The fitting effect of the MAE loss is better compared to that of the MMD loss. The sample points in the middle of the probability distributions based on MMD loss are not exactly matched to the target Gaussian probability distribution and are all higher than the target Gaussian probability distribution. Except for a few sample points, the probability distributions based on MAE loss coincide with the target Gaussian distribution.

Fig. 11 shows the probability distributions after 300 training steps for PBS with $m = 6$. As the number of modes increases, the probability distribution after training still fits well with the target Gaussian distribution for both the MMD loss and the MAE loss. Regardless of the circuit depth, the two sides of the probability distribution after training largely match with the target distribution. But when the circuit depth $d = 4$, the probability distribution after training has a weaker fitting effect with the parameters increasing that the middle of the probability distribution is difficult to reduce in a short training step because of the large initial probability value.

The loss function for different training steps is shown in Fig. 12 where Fig. 12a represents the MMD loss with the Gaussian kernel bandwidth of $\sigma = \frac{\sqrt{2}}{2}$, Fig. 12b means the MAE loss. Since the initial parameters all start with random values, the initial values of the loss function vary greatly for both MMD loss and MAE loss. The MMD loss and the MAE loss both decrease rapidly in the first steps and thereafter converges gradually. After 300 training steps, the MAE loss with mode $m = 4$ and depth $d = 2$ decreases from 4.15×10^{-2} to 6.53×10^{-3}. From Fig. 12b, we can see that the MAE losses finally converge to different values for different modes and depths. The MMD loss with mode $m = 4$ and depth $d = 2$ decreases from 6.00×10^{-3} to 1.08×10^{-5} after 300 training steps. Unlike MAE losses, all the MMD losses generally decrease to the same value for different modes and depths.

Fig. 13 shows the variances between the output and target probability distribution using MMD loss and MAE loss. These two variances decrease rapidly in the first period and gradually converge, like the circuit losses. After 300 training steps, the variances with mode $m = 4$ and depth $d = 2$ using MMD loss and MAE loss decrease from 3.09×10^{-3} and 2.48×10^{-3} to 1.46^{-4} and 1.10×10^{-4}, respectively. As the MMD loss decreases, the variance does not always trend downward, but may fluctuate up and down or increase, e.g., the variances with $m = 4, d = 2$ and $m = 6, d = 2$.

Table 3 shows the parameter numbers for different modes and depth, 10 percent training step, minimum loss, and minimum variance. The 10 percent training step represents the training step when the circuit loss satifies

$$loss \ < \ Min + (Max - Min) \times 10\%, \tag{19}$$

where Min and Max indicate the minimum value and the maximum value of the circuit loss. The circuit learning smoothes out and gradually converges after the 10 percent training step. From Table 3, we can observe that except for MMD loss with $m = 6, d = 4$, the 10 percent training steps

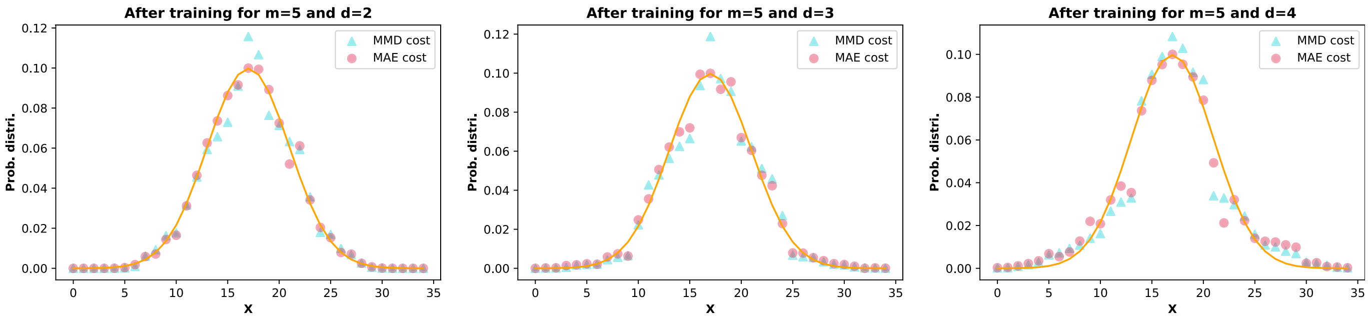


Fig. 10. Scatterplot after training at different depths for $m = 5$.

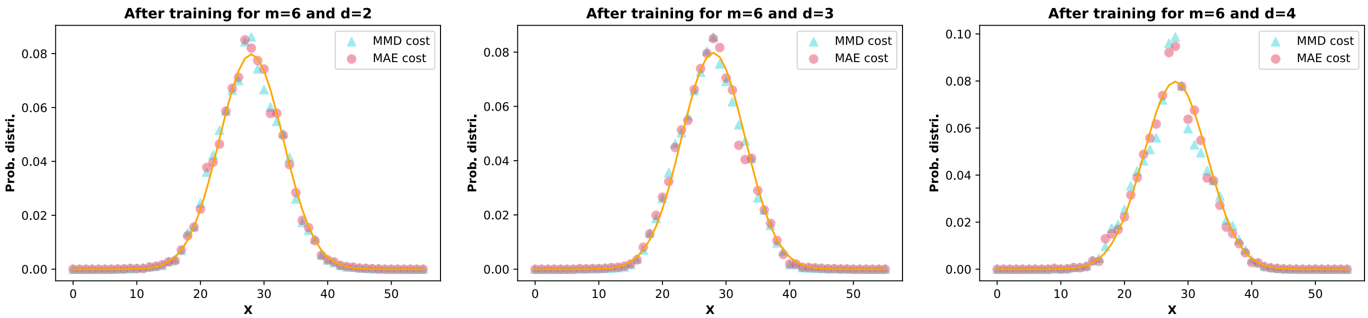
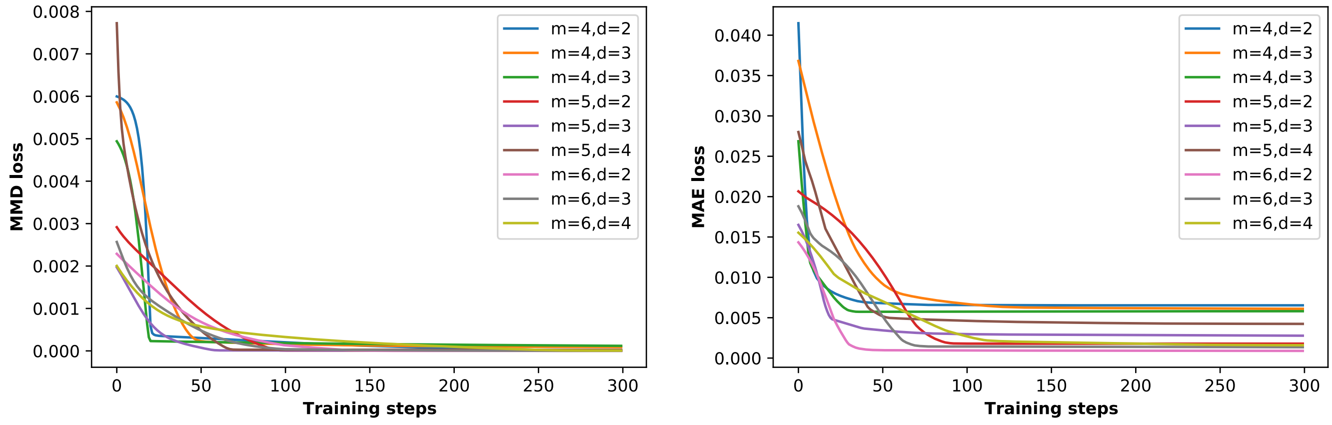


Fig. 11. Scatterplot after training at different depths for $m = 6$.



(a) MMD loss (b) MAE loss

Fig. 12. The circuit losses for the PBS model.

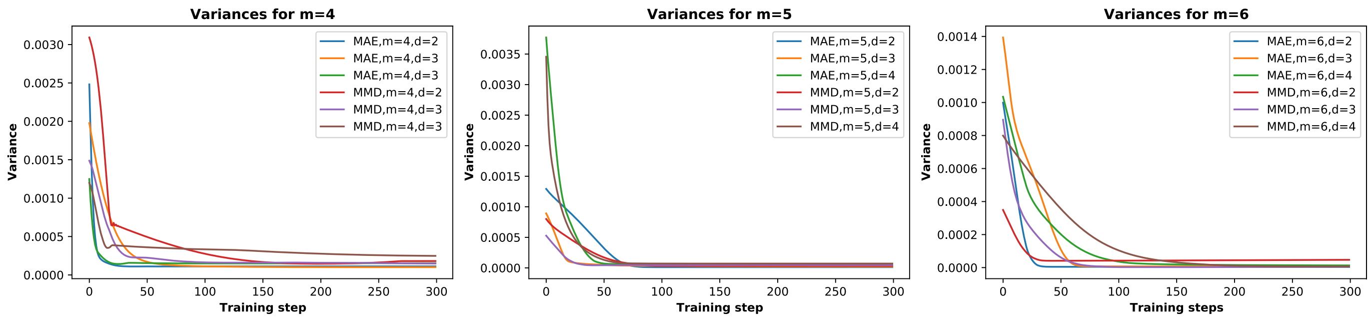


Fig. 13. The variances using MMD loss and MAE loss.

TABLE 3
The Experimental Comparison Using Two Losses

loss	mode	depth	parameter number	10% training step[1]	Minimum loss	Minimum variance[2]
	4	2	20	20	1.08×10^{-5}	1.46×10^{-4}
	4	3	30	39	9.08×10^{-5}	1.48×10^{-4}
	4	4	40	19	1.19×10^{-4}	2.48×10^{-4}
	5	2	26	76	6.85×10^{-6}	3.20×10^{-5}
MMD	5	3	39	38	5.36×10^{-6}	4.20×10^{-5}
	5	4	52	42	2.52×10^{-5}	6.99×10^{-5}
	6	2	32	86	4.52×10^{-7}	4.13×10^{-5}
	6	3	48	65	8.14×10^{-7}	$\mathbf{2.54 \times 10^{-6}}$
	6	4	64	143	7.17×10^{-6}	6.72×10^{-6}
	4	2	20	11	6.53×10^{-3}	1.10×10^{-4}
	4	3	30	51	6.10×10^{-3}	1.01×10^{-4}
	4	4	40	20	5.72×10^{-3}	1.42×10^{-4}
	5	2	26	72	1.78×10^{-3}	1.11×10^{-5}
MAE	5	3	39	31	2.77×10^{-3}	1.99×10^{-5}
	5	4	52	41	4.23×10^{-3}	5.24×10^{-5}
	6	2	32	29	8.90×10^{-4}	4.19×10^{-5}
	6	3	48	58	1.35×10^{-3}	$\mathbf{6.91 \times 10^{-6}}$
	6	4	64	95	1.59×10^{-3}	1.32×10^{-5}

[1]*10 percent training step refer to the training step when the circuit loss less than 10 percent between maximum and minimum of the loss function, e.g., the training step when Eq. (19) is satisfied.*
[2]*variance refer to variance of the probability distribution and the target probability distribution when trained with different loss functions.*
[3]*The bolded data indicate the minimum variance of the corresponding loss function respectively.*

for all the circuit losses satifying Eq. (19) are less than 100. Regardless of the number of modes and circuit depths, all the MAE losses after 300 training steps are decreased to between 6.53×10^{-3} to 8.90×10^{-4} from Fig. 12b and Table 3. All the MMD losses after 300 training steps are decreased to between 1.19×10^{-4} to 4.52×10^{-7}.

In the experimental process, we found that the function fitting effect using the MAE loss was better and more sensitive than that using the MMD loss in the previous steps of the rapid decline. The fitting effect using MMD loss is a gradual process, getting better as the quantum circuit is trained. As circuit learning continues, the output probability distribution fits the target Gaussian distribution gradually, and the circuit learning converges after 300 training steps. Different from MMD loss, the MAE loss is dependent on the sample points location of the probability distribution, the probability distribution with $m = 4, d = 2$ after training the first 10 steps has been fitted well to the target Gaussian distribution. The effect of MAE loss on function fitting is more visual than that of MMD loss.

There are still some flaws in the experimental simulation. The total probability of the target probability distribution for the mode $m = 4$ and $m = 6$ is less than 1. Since the Gaussian distribution is symmetrical and the sample size of the output probability distribution is even, the target Gaussian distribution generated by the Scipy library discards the probability of the last sample point. Moreover, we can exclude the first column parameters from the training process to speed up the computation of the numerical simulation in updating the circuit parameters, since the first column rotation gates (phase shifters) have no effect on the probabilities of all possible output configurations. From the experimental results, the PBS circuit is still able to generate the probability distribution of corresponding sample points well. In terms of generating sample space, our scheme using m modes and n photons can be scaled as $\binom{m+n-1}{n}$, which is the super-exponential of n, compared to the classical one.

5 Conclusion

We have argued that besides providing solid examples of quantum computational advantage, BS has the potential to solve practical application problems by fitting a Gaussian function. We introduce a parameterized structure into BS and propose a PBS model. The proposed fixed structure can reduce the model complexity as the circuit depth d grows. We develop a quantum circuit learning scheme based on the PBS model to implement Gaussian function fitting. In this hybrid quantum-classical approach, the PBS model can be used to achieve quantum state preparation, quantum circuit execution, and measurement. The circuit parameters are updated by evaluating the loss function on classical computers. Our results have shown that the PBS circuit can implement fast-fitting of Gaussian function. This fact provides extra evidence for challenging the common belief that BS is difficult to solve practical problems, which deserves more attention in the future.

Acknowledgments

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Jinjing Shi received the BS and PhD degrees from the School of Information Science and Engineering, Central South University, Changsha, China, in 2008 and 2013, respectively. She is now an associate professor with the School of Computer Science and Engineering, Central South University, China. She was selected in the "Shenghua lieying" talent program of Central South University, China, and Special Foundation for Distinguished Young Scientists of Changsha, in 2013 and 2019, respectively. Her research interests include quantum computation and quantum cryptography. She has presided over the National Natural Science Foundation Project of China and that of Hunan Province. There are 50 academic papers published in important international academic journals and conferences. She has received the second prize of natural science and the outstanding doctoral dissertation of Hunan Province, in 2015, and she has received the Best Paper Award in the International Academic Conference MSPT2011 and Outstanding Paper Award in IEEE ICACT2012.

Yongze Tang received the BS degree from the School of Information Engineering, Jiangxi University of Science and Technology, China, in 2019. He is currently working toward the master's degree in the School of Computer Science and Engineering, Central South University, Changsha, China. His research interests mainly include quantum computing, boson sampling, and quantum neural networks.

Yuhu Lu received the BS degree in computer science and technology from Huaqiao University, China, in 2017. He is currently working toward the master's degree in the School of Computer Science and Engineering, Central South University, Hunan, China. His research interests mainly include quantum cryptography, quantum computing, and quantum neural networks.

Yanyan Feng received the BS degree from Henan University, China, and the MS degree from Central South University, China. Currently, she is currently working toward the PhD degree in the School of Computer Science and Engineering, Central South University, Changsha, China. Her research interests mainly include quantum communication, quantum circuit models and their applications, and quantum circuit learning supremacy.

Ronghua Shi received the BS, MS, and PhD degrees in computer application technology from Central South University (CSU), China, in 1986, 1989, and 2007, respectively. He is the supervisor of PhD students, and the team leader of the "Communication System and Network Security" Group in the School of Computer Science and Engineering, Central South University, Changsha, China. He is the executive director of Railways Specialty Committee, the chairman of Hunan Internet of Things Committee, the vice-chairman of the Hunan Higher Education Computer Society Professional Committee, and the executive director of Provincial Communication Society as well. He has received the State Council Special Allowance. His professional field covers computer science and technology, information and communication engineering, etc., and his research interests include network and information security, quantum cryptography, quantum secure communications, etc. His teaching curriculums contain Special Topics of information security technology, modern cryptography theory, and application, computer network communications, introductions to information science, etc. In recent years, he has hosted more than 10 research projects such as the National 863 Project, the Natural Science Foundation Project, the Ministry of Education Doctoral Fund Project, etc. He has received two Provincial and Ministerial Appraisal projects, two projects for the Second Prize of Provincial Science and Technology Progress and one project for that of Third Prize, and two articles for the Second Prize of Outstanding Papers of Natural Science in Hunan Province. He has published more than 80 articles in domestic and foreign academic journals, which include about 40 SCI or EI articles.

Shichao Zhang (Senior Member, IEEE) received the PhD degree in computer science from Deakin University, Australia. He is currently a China National-level-Title professor with the School of Computer Science and Technology, Central South University, China. His research interests include information quality and pattern discovery. He has published about 90 international journal papers and more than 100 international conference papers. He has won 16 national-class grants, and eight China provincial/ministerial Awards. He served/is serving as an associate editor for the *ACM Transactions on Knowledge Discovery from Data*, *IEEE Transactions on Knowledge and Data Engineering*, *Knowledge and Information Systems*, and *IEEE Intelligent Informatics Bulletin*, as conference chair, PC chair and vice PC chair for 10 more international conferences. He is a member of the ACM.